

Chapter 2: Fundamentals of a Computer System

| Learning Objectives

- Identify features and classifications of computers.
- Understand the functions and devices within a computer system.
- Identify basic components, ports, and networking concepts.
- Differentiate between data transmission modes and media.

| 2.1 What is a Computer?

A computer is an electronic device that accepts or collects data, processes it according to given instructions, and produces the desired output.

Key Features

- **Speed:** Executes millions of instructions per second.
- **Accuracy:** Provides correct output if given correct data and instructions.
- **Efficiency:** Works continuously without fatigue.
- **Versatility:** Can perform multiple tasks simultaneously.
- **Storage & Retrieval:** Stores large amounts of data in a small space for quick retrieval.

| 2.2 Classification of Computers

2.2.1 By Size

- **Super Computers:** Highest computing power, large, expensive. Used for scientific/military purposes (e.g., NASA).
- **Mainframe Computers:** Less powerful than Super computers. Used in large businesses where multiple users connect via terminals.
- **Mini Computers:** Slower and less expensive than mainframes. Used in banking for sharing data across a few terminals.
- **Personal Computers (Micro):** Small, portable, and meant for individual use (e.g., Laptops, Desktop PCs).

2.2.2 By Technology

- **Analog Computers:** Use analog signals (speed, pressure, temperature). Examples: Speedometers, sensors.
- **Digital Computers:** Use digital signals (electricity). Used in everyday computing.
- **Hybrid Computers:** Combination of analog and digital. Example: ECG machines that capture analog heartbeats and convert them to digital signals.

2.3 Functions of a Computer System

The core workflow involves: **Input** → **Processing** → **Output**, with **Storage** and **Control** integrated throughout.

- Data and instructions enter via Input devices and are stored in Primary Memory.
- The CPU uses this data for processing.
- Processed information is returned to Primary Memory and then sent to Output devices.
- Data can be sent to Secondary Storage for permanent keeping.
- The **Control Unit (CU)** manages signals across all devices.

2.4 Basic Components

2.4.1 Input Devices

- **Keyboard:** Most common; 101 to 108 keys.
- **Pointing Devices:** Mouse, Touch Pad, Touch Screen, Joy Stick, Light Pen.
- **Imaging/Video:** Digital Camera, Webcam, CCTV.
- **Scanners:** Flatbed, Bar Code Reader, MICR (Banks/Cheques), OCR (Text editing), OMR (MCQ marking).
- **Other:** ATM (card readers), Microphone (sound).

2.4.2 Output Devices

- **Soft Copy:** Monitors (CRT - large/hot; LCD - flat; LED - bright/efficient) and Multimedia Projectors.
- **Hard Copy:**
 - **Impact Printers:** Dot Matrix (noisy, uses ribbons), Line Printer (very fast).
 - **Non-Impact Printers:** Laser (high quality, toner), Ink Jet (liquid ink), Thermal (heat-based receipts), Plotter (architectural plans).
- **Sound:** Speakers.

2.4.3 Central Processing Unit (CPU)

- **ALU (Arithmetic & Logical Unit):** Performs calculations and logic.
- **CU (Control Unit):** Controls all system devices.
- **Memory Registers:** Temporary internal storage for the ALU.

Speed: Measured in Hertz (Hz), Megahertz (MHz), and Gigahertz (GHz).

2.4.4 Computer Memory

Primary Memory (Main Memory)

- **RAM (Random Access Memory):** Volatile; stores temporary data for input/output.
- **ROM (Read Only Memory):** Non-volatile; stores BIOS (boot instructions).
- **Cache Memory:** Fastest; acts as a mediator between CPU and RAM.

Secondary Memory (External Storage)

- **Magnetic:** Hard Disk (Platters, Spindle, Head, Actuator), Magnetic Tape (backups).
- **Optical:** CD (650-900MB), DVD (4.7-9.4GB), Blu-ray (25-128GB). Variations include ROM, R (Write Once), and RW/RE (Rewritable).
- **Solid State:** USB Flash Drives, Memory Cards (SD cards). No moving parts.

2.5 Computer Ports

- **PS/2:** Purple (Keyboard), Green (Mouse).
- **Parallel:** 25 holes; used for older printers.
- **HDMI:** High-definition interface for monitors/projectors/TVs.
- **RJ 45:** Networking port for routers/switches.
- **Audio Jacks:** Blue (Line In), Green (Out/Headphones), Pink (Mic).
- **USB:** Universal port for most devices.
- **VGA (Video port):** 15 holes; connects monitors/projectors.
- **Serial:** Connects older modems.

2.6 Data Communication

The transmission of data between two or more devices/computers.

Components

- **Sender:** The source of data.
- **Medium:** The path for transmission.
- **Receiver:** The destination.

Transmission Modes

- **Simplex:** One way only (e.g., Radio, TV).
- **Half-duplex:** Both ways, but one at a time (e.g., Walkie-talkies).
- **Full-duplex:** Both ways simultaneously (e.g., Telephone).

Transmission Media

- **Guided (Wired):** Twisted Pair (UTP/STP), Coaxial Cable, Fiber Optics (glass/light).
- **Unguided (Wireless):** Radio waves (Wi-Fi, Bluetooth), Microwaves (Satellite), Infrared (Remotes).

2.7 Computer Networking

Devices

- **NIC:** Connects computer to a network.
- **Switch/Hub:** Mediators. Switch is faster/smarter (relevant computer only); Hub sends to all.
- **Router (Gateway):** Connects different networks; finds best path.
- **Modem:** Translates Analog to Digital (Modulation/Demodulation).
- **Firewall:** Security barrier (Hardware/Software) between internet and network.

Types and Topologies

- **Types:** LAN (Local/Building), MAN (Metropolitan/City), WAN (Wide/Internet).
- **Topologies:** Star (central hub), Bus (main cable backbone), Ring (circular), Tree (Bus + Star), Mesh (all interconnected).

Summary

A computer is a versatile system defined by speed and accuracy. It is classified by size and technology. Its components include various input/output devices, a CPU for processing, and multi-tiered memory (RAM, ROM, Cache, Secondary). Communication relies on specific modes and media, and networking allows for resource sharing (printers, software, storage) and security via firewalls. While networking offers vast benefits like e-mail and central control, it also presents challenges like viruses and hacking risks.